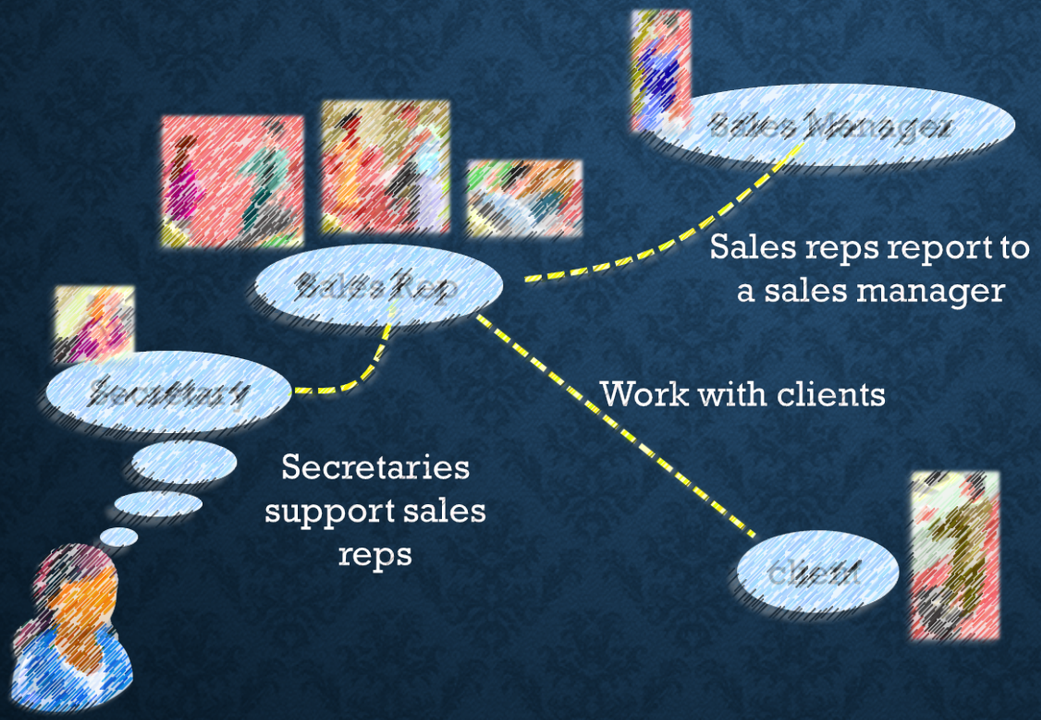
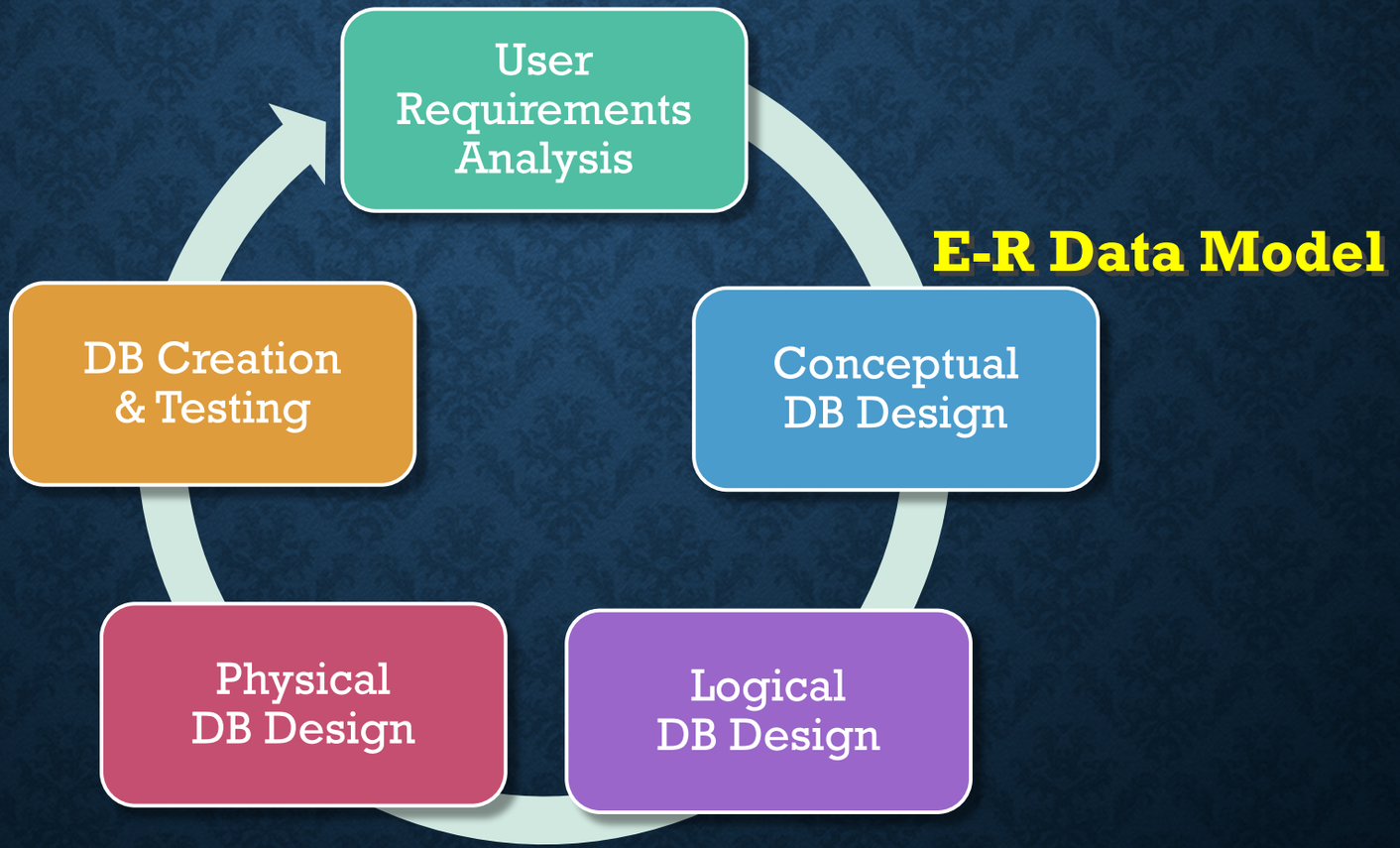
Relational Database & SQL

Entity-Relationship Modeling



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Computer Information Systems, Georgia State University

A database design life-cycle



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Entity-Relationship (ER) modeling



Conceptual Ideas

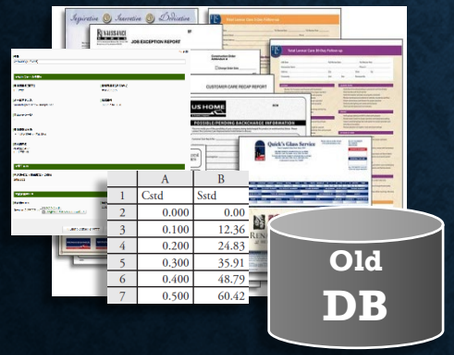


Architecture drawing

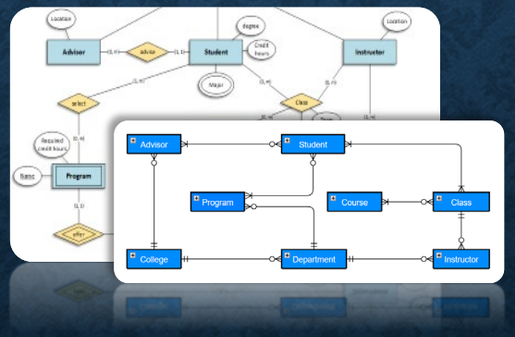


Construction

User Requirement Analysis



ER Modeling



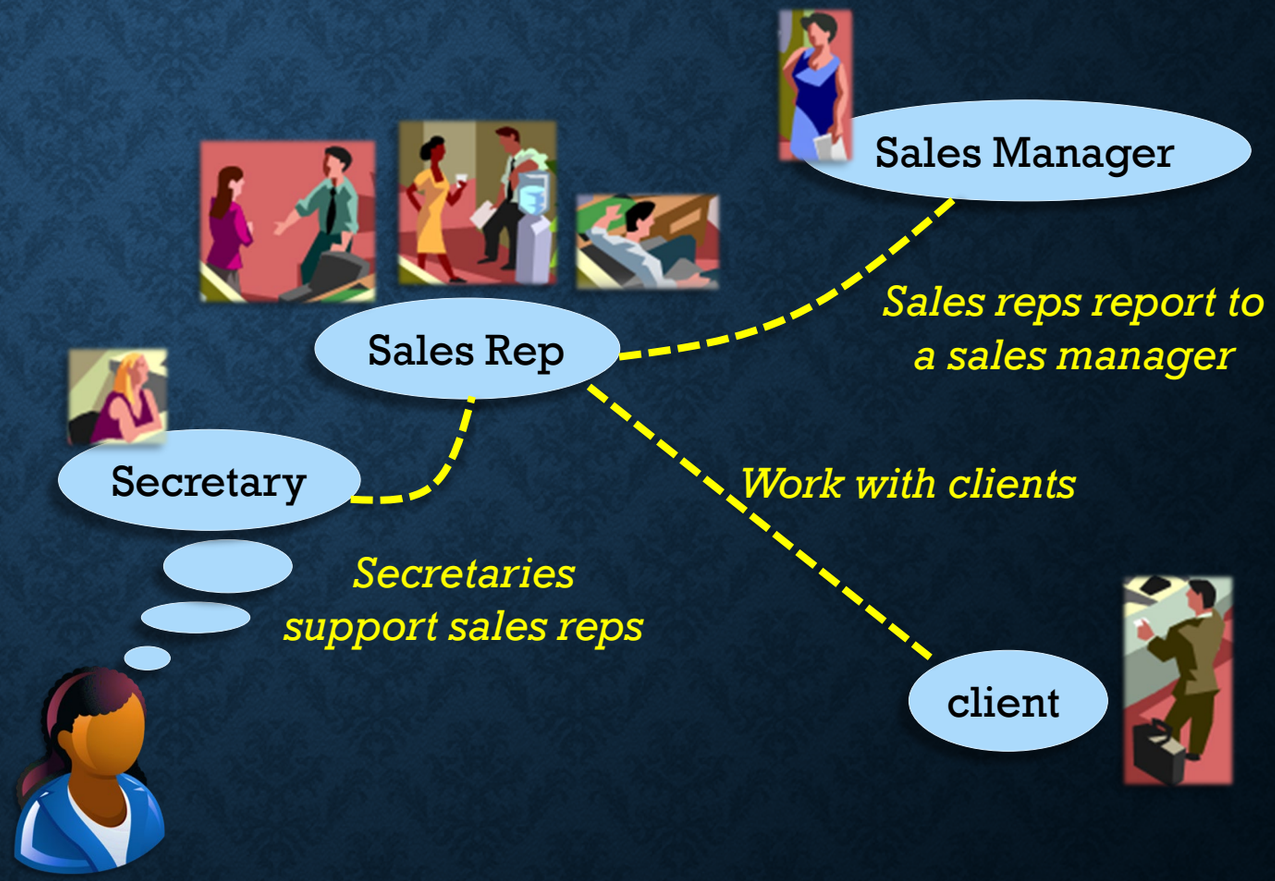
Database Creation



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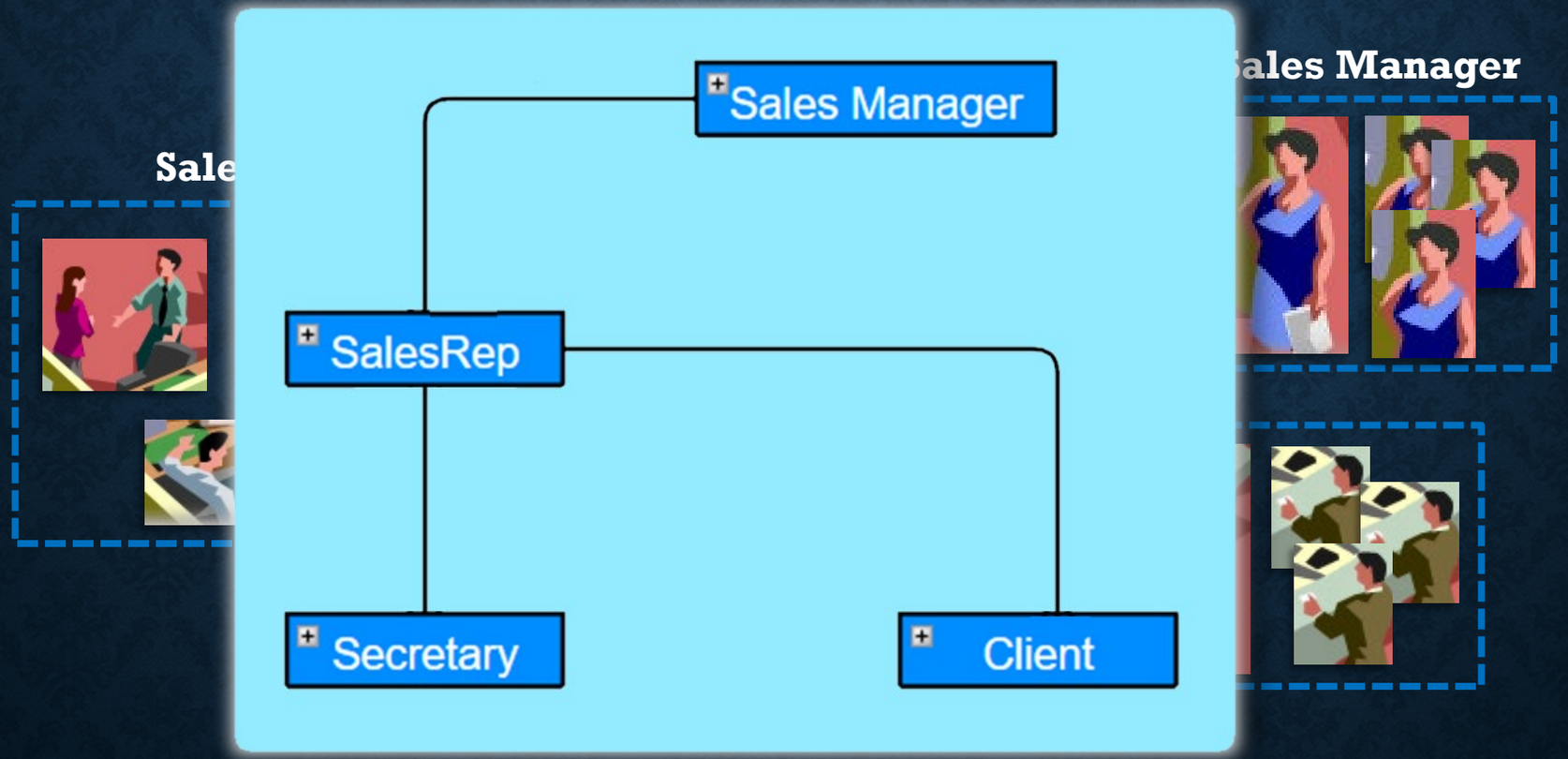
Discover & abstract the business world as entities and relationships





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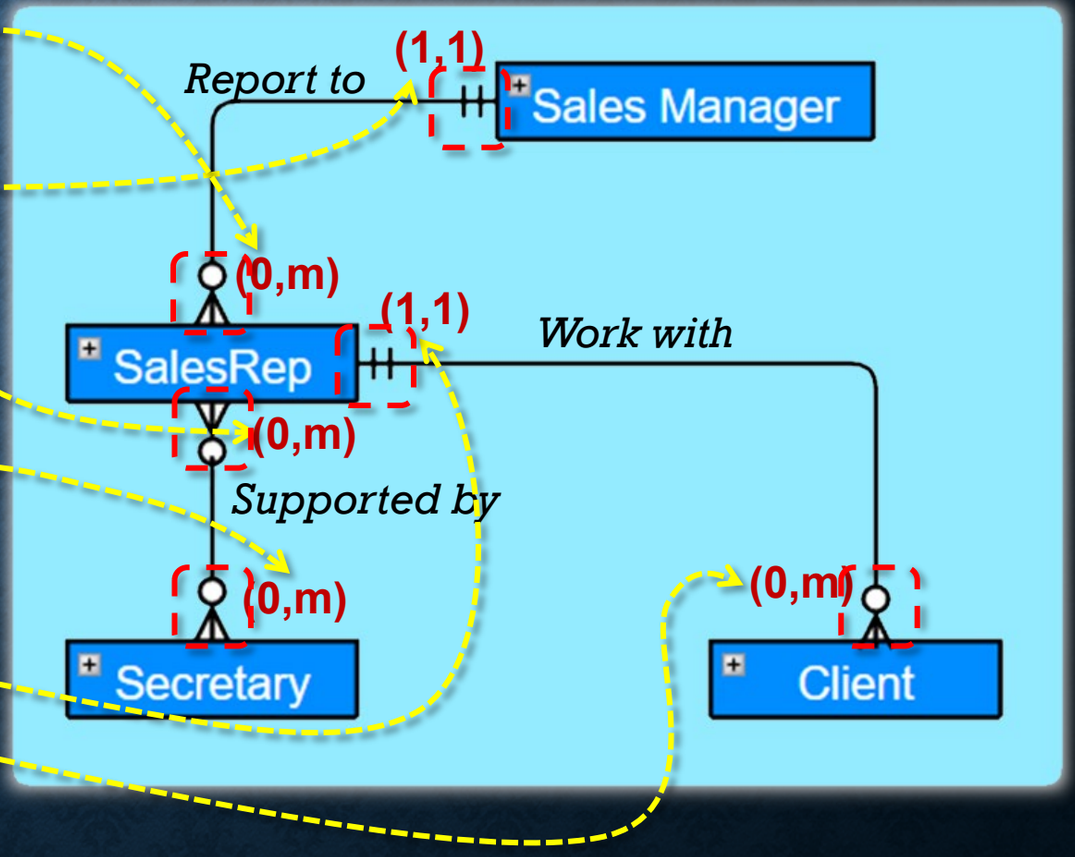
Represent entities & relationships in an ER model



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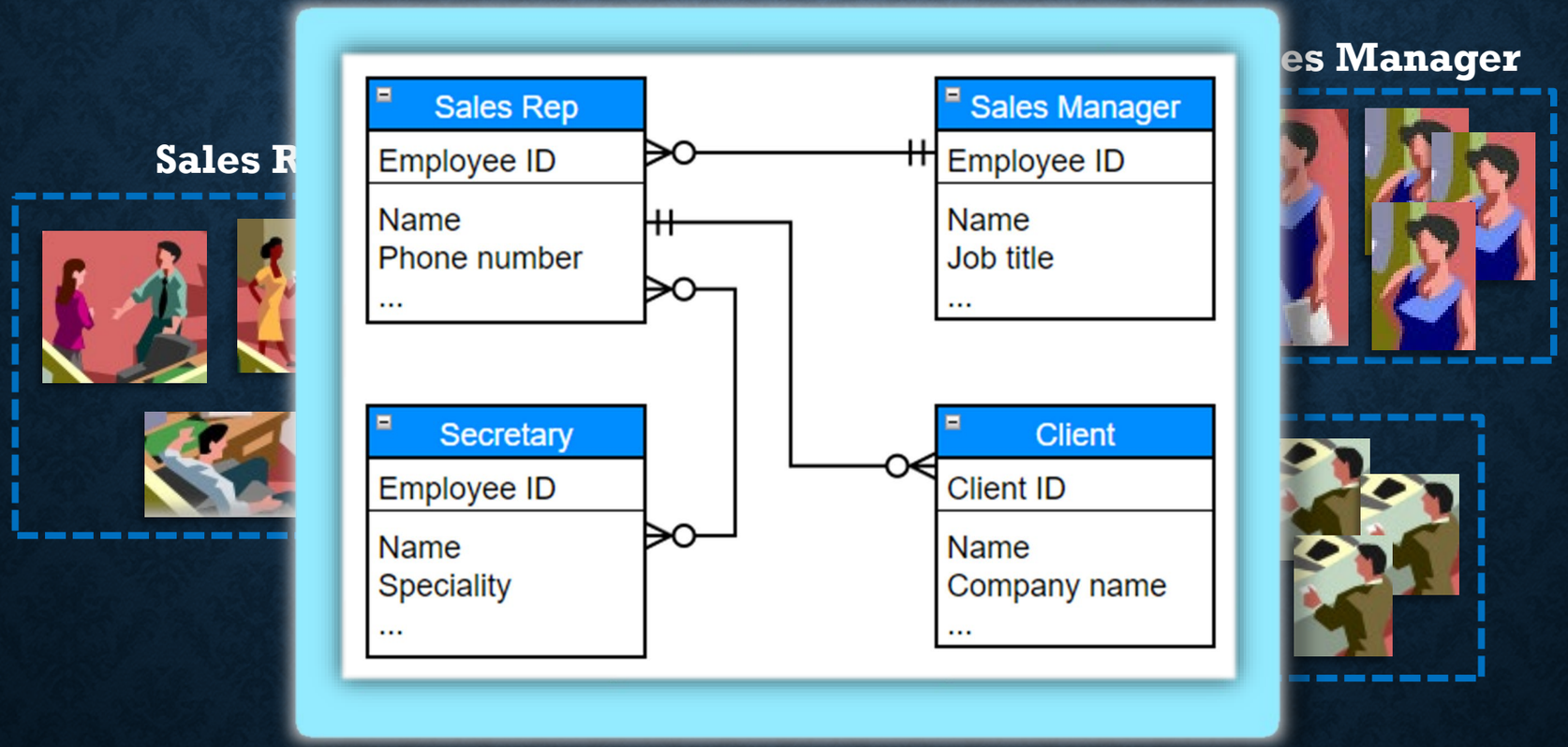
Present business rules as relationship cardinalities in the ER model

1. A manager may supervise none or many sales reps
2. Each sales rep always reports to one and only one manager
3. A secretary may support none or multiple sales reps
4. A sales rep may have one or many supporting secretaries, or none in temporary situations
5. A client always has a dedicated sales rep
6. A sales rep may have none or many clients

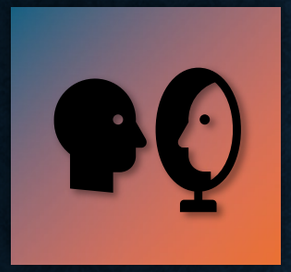


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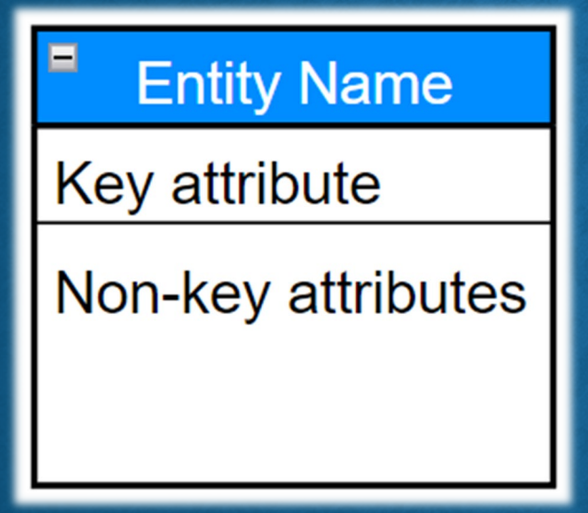
Identify attributes (fields) of entities

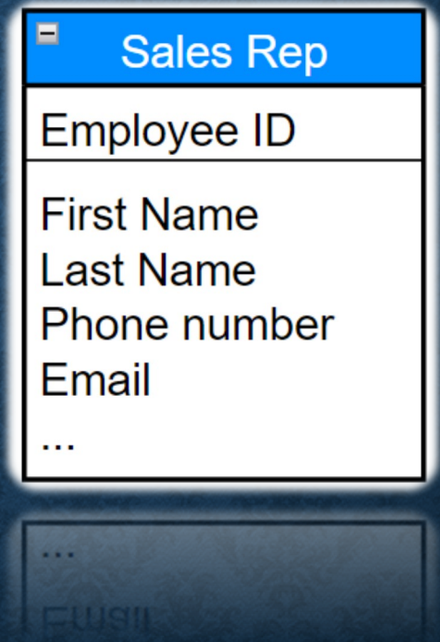


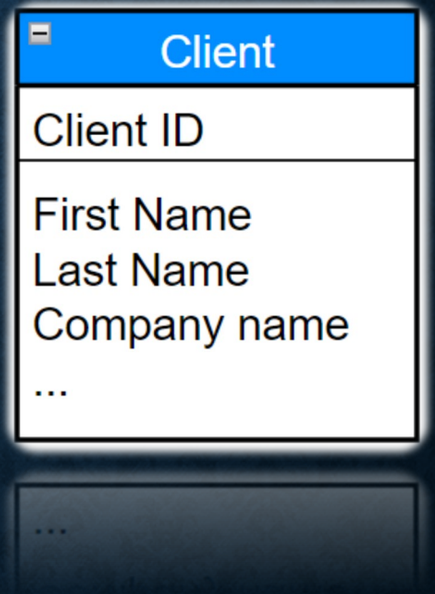
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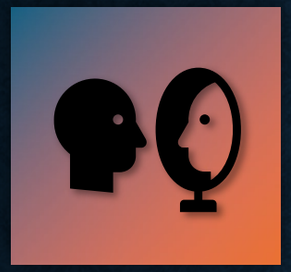
Entity Notations



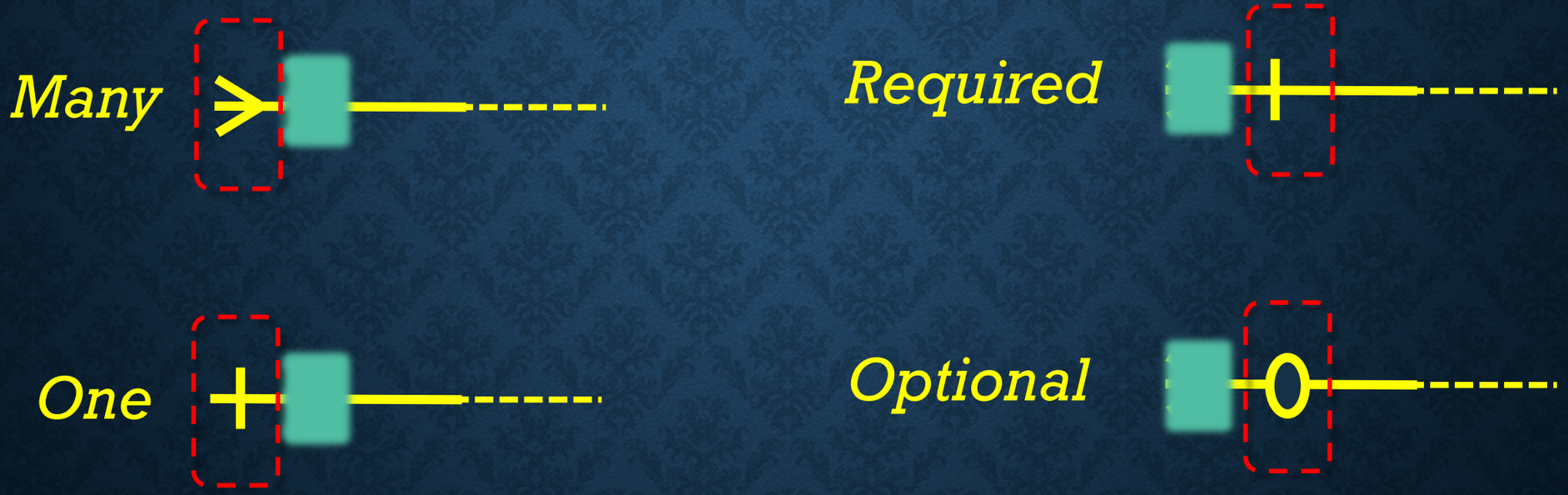




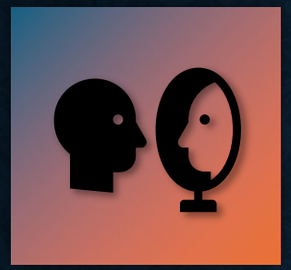
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Relationship notations



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Explore ER modeling & notations

GenAI

Work with your AI tool to

- Understand the role of ER modeling in database design
- Explore the Claw's Foot notations and examples
- Compare and contrast different ER modeling notations